Dot products and weighted sums

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Dot products and weighted sums before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Representing inputs as vectors

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

A small model can combine several measurements into one **score**. Let x = [4, 2] contain fictional width and height measurements in cm. Set **weight**s w = [2, -1].

Multiply matching positions, then add: s = (2 times 4) + (-1 times 2) = 6. The letter x names the **input** vector.

We call this calculation a **dot product**, written w dot x. The vectors must have the same number of components. Each **weight** belongs to one named feature.

These hand-selected **weight**s demonstrate a computation; no **training** has happened. The result 6 is a **score**, not a probability or a claim that this parcel is safe.

Feature	Weight times input	Contribution
Width	2 times 4	8
Height	-1 times 2	-2
Total	8 plus -2	6

The coefficients carry **score**-per-cm units in this example, making both **contribution**s compatible **score** units. A **weight**ed sum needs a consistent definition of units.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Dot products and weighted sums trace and record the first step you can explain without help.

Use W2: Explain cancellation for **Dot product**s and **weight**ed sums. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Dot products and weighted sums explanation against the supplied facts before discussing it.

Use the worked case for **Dot product**s and **weight**ed sums. Proposed explanation: “Multiply the sums instead”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Compute a new dot product independently

Use x = [3, 1, 2] and w = [-2, 4, 0.5]. Give each **contribution** and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-**weight** list is invalid for this **input**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Calculate contributions

> Calculate contributions. Apply the supplied Dot products and weighted sums facts and keep your reasoning.

For w = [2, -1] and x = [3, 5], write both **contribution**s and their sum.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Explain cancellation

> Explain cancellation. Apply the supplied Dot products and weighted sums facts and keep your reasoning.

For w = [2, -1] and x = [2, 4], explain why **score** zero does not mean the **input** contains no information.

My reasoning and evidence

W3 Preserve a unit change

> Preserve a unit change. Apply the supplied Dot products and weighted sums facts and keep your reasoning.

Width stays in cm and height changes to mm. For the worked parcel [4 cm, 20 mm], give the revised **weight**s and **score**.

My reasoning and evidence

W4 Compute a new dot product independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Compute a new dot product independently. Apply the supplied Dot products and weighted sums facts and keep your reasoning.

Use x = [3, 1, 2] and w = [-2, 4, 0.5]. Give each **contribution** and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-**weight** list is invalid for this **input**.

> **My reasoning and evidence**

W5 Find lost information

> Find lost information. Apply the supplied Dot products and weighted sums facts and keep your reasoning.

> Verify that [4, 2] and [5, 4] both **score** 6 with w = [2, -1]. Can a later step receiving only this **score** distinguish those two records?

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Dot products and weighted sums to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled **dot product**s and **weight**ed sums **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate a **dot product** by matching feature positions.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Dot products and weighted sums

> Draw each input's contribution to a toy weighted brightness score. Explain the labelled step before checking the changed condition.

Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**.

Inputs [2, 5]; **weight**s [3,-1]; no bias. **Score** is 2×3+5×(-1).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/2

Weighted brightness score

Input 2 x weight 3 = 6

Input 5 x weight -1 = -5

Add the products and bias 0: total = 1

Keep the total (identity) -> output 1

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/2

Dot products and weighted sums

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Multiply first input

Left contribution=2 x 3=6.

2. Multiply second input

Right contribution=5 x (-1)=-5.

3. Add contributions

Weighted sum=6-5=1; this is a score, not a probability.

CHANGE ONE THING

Increase only the right input from 5 to6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which connection explains why more right brightness lowers this **score**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Weight**ed sum=6-5=1; this is a **score**, not a probability.

1. Multiply second **input**

2. Add **contribution**s

3. Multiply first **input**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Add **contribution**s **Weight**ed sum=6-5=1; this is a **score**, not a probability. A **weight** controls how strongly an **input** contributes to the **score**.

The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each **contribution** before adding them.

This **dot product** is a signed **score**; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Increase only the right **input** from 5 to6. Which conclusion is supported by the supplied facts?

1. The **score** becomes 6-6=0.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The **score** becomes 6-6=0. The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

Repair a missing building block

Review Representing **input**s as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical **input**s?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Dot products and weighted sums picture from its labels, then record a relevant limit.

01. Dot products and weighted sums: the case

AI ACADEMY / AI-5/2 · 01 · LOOK AT THE FACTS

Dot products and weighted sums: the case

The situation

Draw each input's contribution to a toy weighted brightness score.

Supplied evidence

Inputs [2, 5]; weights [3,-1]; no bias. Score is 2×3+5×(-1).
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain dot products and weighted sums using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

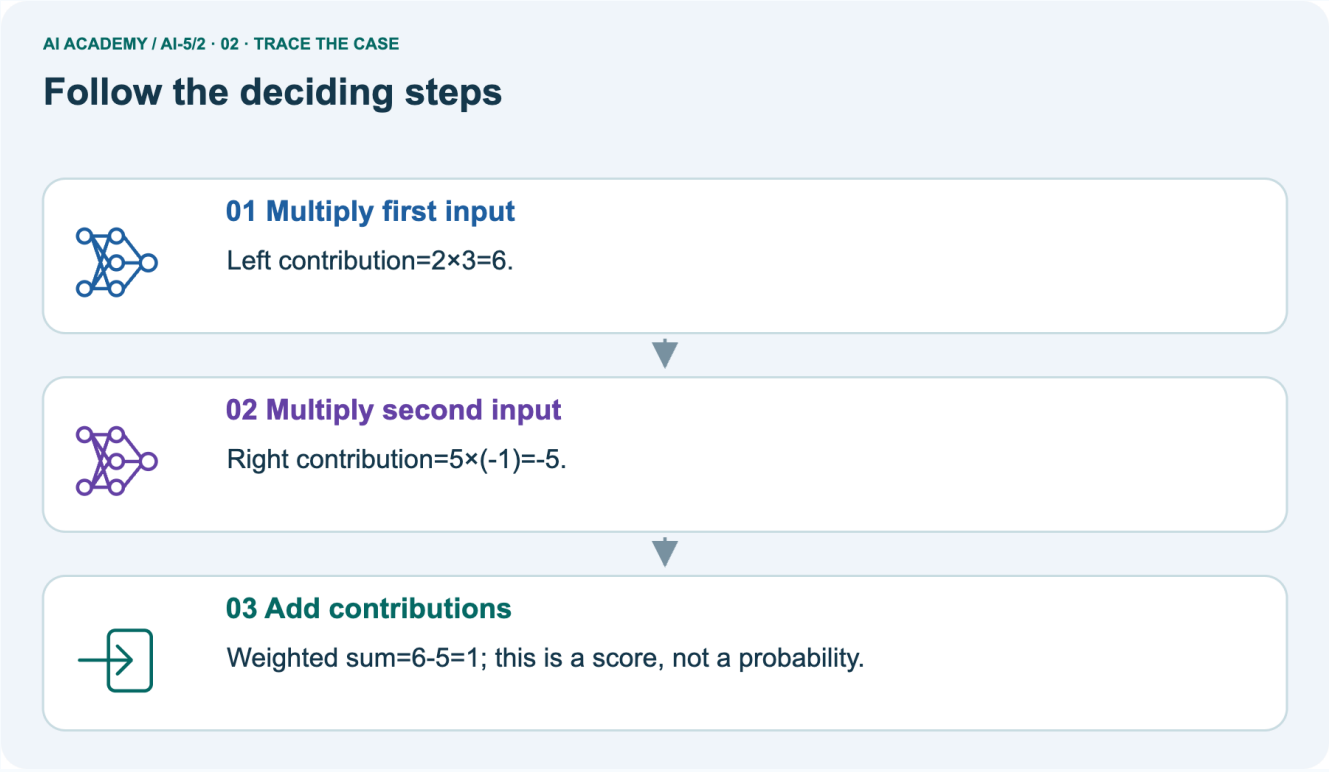
Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**. Supplied facts: **Input**s [2, 5]; **weight**s [3,-1]; no bias. **Score** is 2×3+5×(-1).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



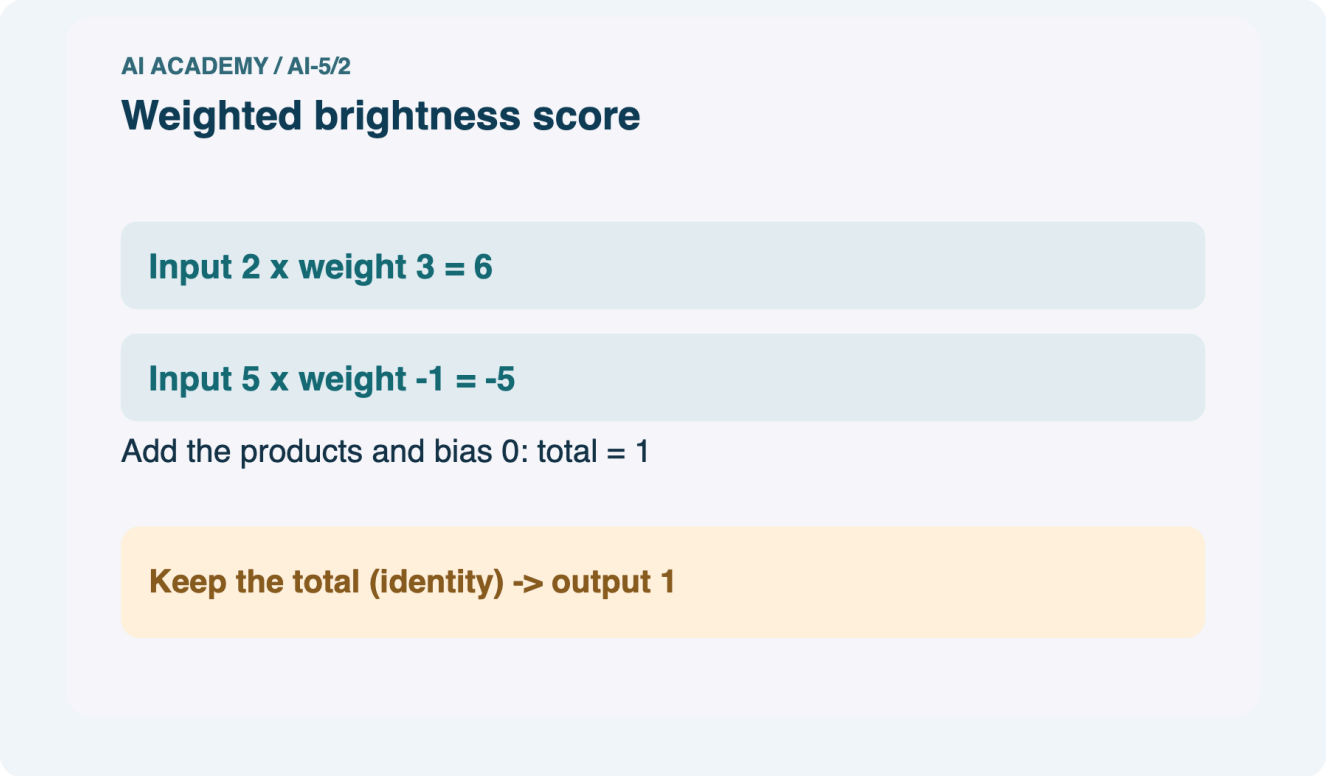
Follow the labelled picture with the matching explanation. Open the vector diagram

Multiply first **input**: Left **contribution**=2×3=6. → Multiply second **input**: Right **contribution**=5×(-1)=-5. → Add **contribution**s: **Weight**ed sum=6-5=1; this is a **score**, not a probability.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Weighted brightness score



Follow the labelled picture with the matching explanation. Open the vector diagram

Weighted brightness **score**. **Input**s: [2, 5]; **weight**s: [3, -1]; bias: 0; activation: identity; **output**: 1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.

AI ACADEMY / AI-5/2 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Weighted sum=6-5=1; this is a score, not a probability.

Changed condition

Increase only the right input from 5 to6.

Worked changed-case result

The score becomes 6-6=0.
Why: The right weight is-1, so adding 1 to that input lowers the score by1.

Follow the labelled picture with the matching explanation. Open the vector diagram

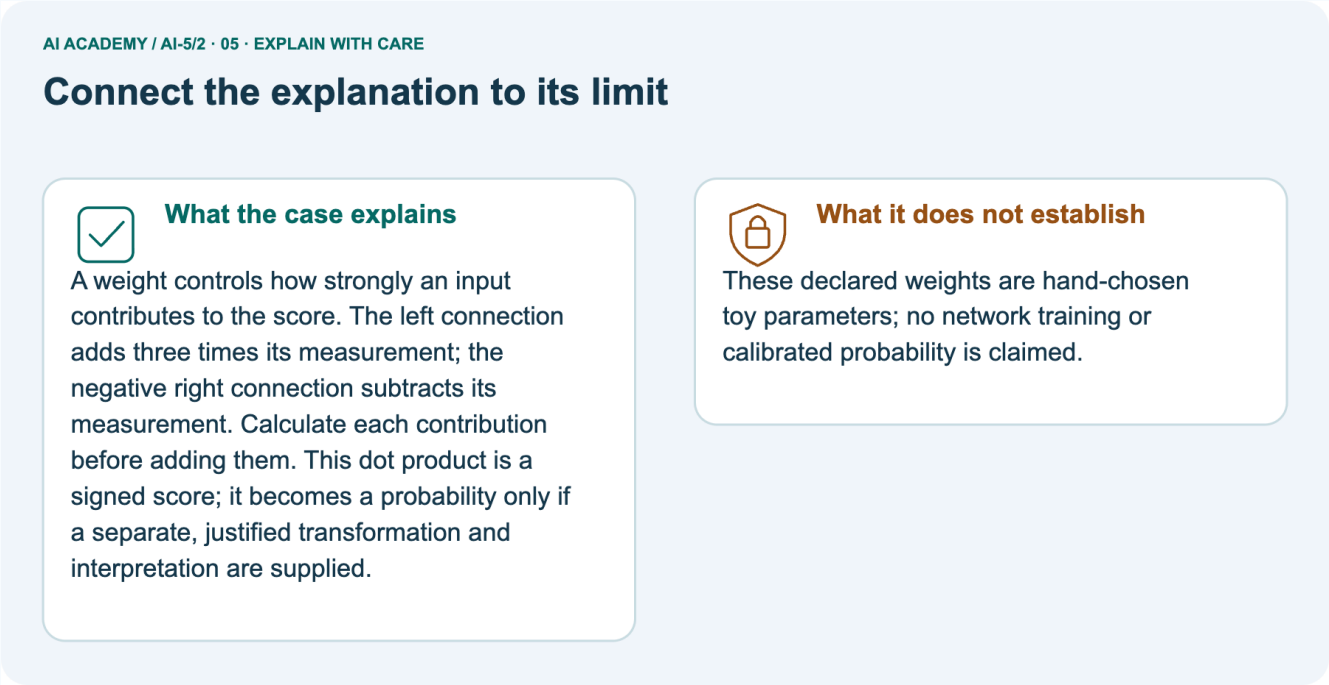
Original case: **Weight**ed sum=6-5=1; this is a **score**, not a probability. Changed condition: Increase only the right **input** from 5 to6. Expected result: The **score** becomes 6-6=0.

Reason: The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **weight** controls how strongly an **input** contributes to the **score**. The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each **contribution** before adding them.

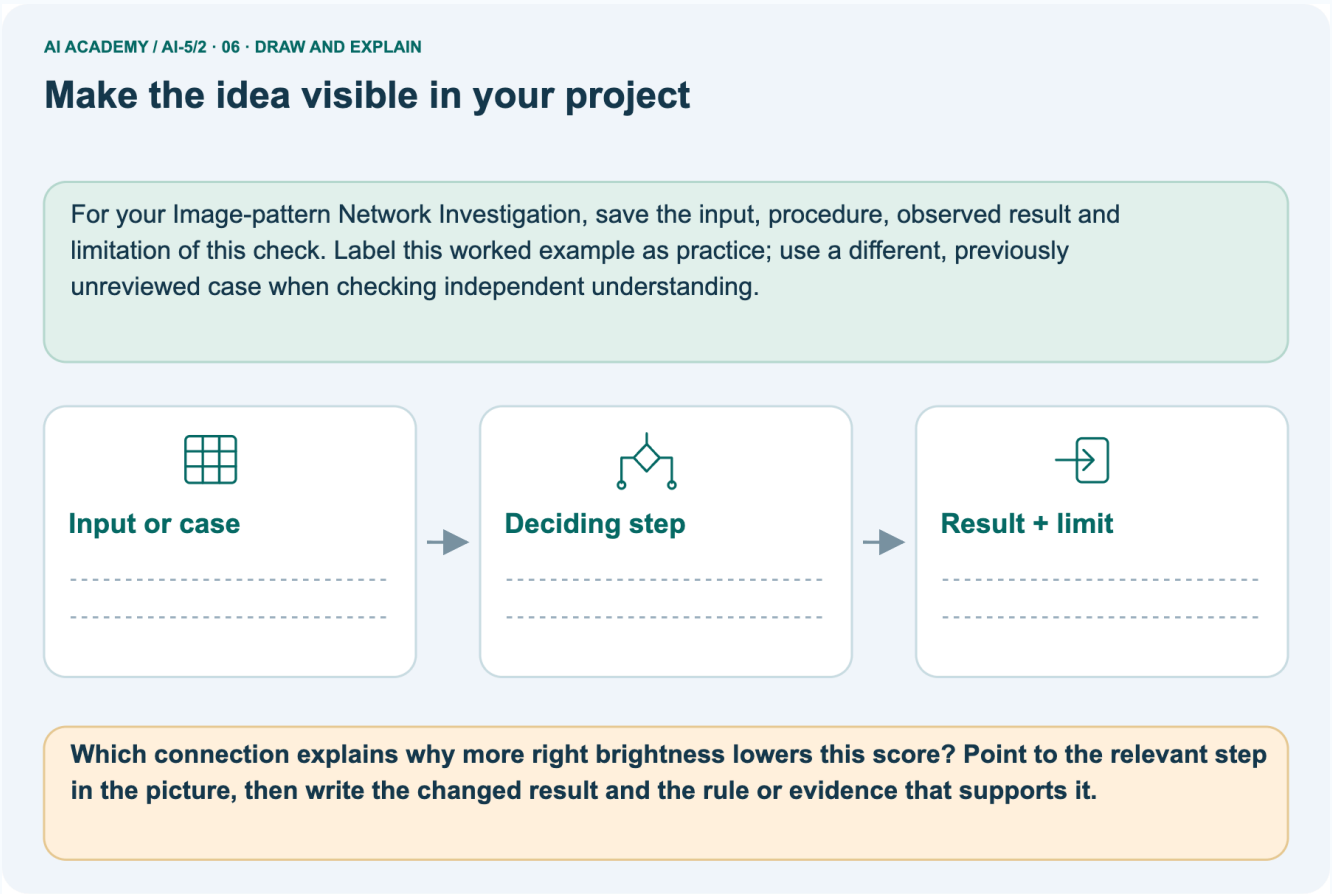
This **dot product** is a signed **score**; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

Limit: These declared **weight**s are hand-chosen toy parameters; no network **training** or calibrated probability is claimed.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which connection explains why more right brightness lowers this **score**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.